

Multi-task recognition motor imagery based on deep learning

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Abstract—Brain-computer interface (BCI) is a new type of human-computer interaction, and eeg signal recognition is the core part of BCI system. In view of the difficulties in feature extraction and low recognition accuracy in the process of multi-class motion image EEG recognition, this paper proposes a multi-class EEG recognition method based on deep learning. The Channel Attention Mechanism (CAM) in deep learning has the advantage of utilizing spatial information, CAM combined with One Versus Rest Filter Bank Common Spatial Pattern (OVRFB-CSP) was used for feature extraction and Softmax classifier was used for classification. This method was applied to the 2008 BCI Competition IV 2a dataset. Recognition of four types of motor imagery tasks (left hand, right hand, foot and tongue). The experimental results showed that the accuracy rate of CAM-OVRFB-CSP reached the highest 83% and the average reached 68.4%, which proved the effectiveness of this method and provided a new idea for the follow-up research.

Keyword: motor imagery, OVRFB-CSP, CAM

I. INTRODUCTION

Brain-computer Interface (BCI) builds a direct information transmission pathway between the brain and the outside world by decoding the information of human brain neural activity during thinking. Electroencephalography (EEG) equipment is the most widely used in BCI, thanks to its portability, low cost, convenient operation, and wide range of participants[1]. Because the key to the realization of BCI technology is the high recognition rate of EEG signals, the extraction of EEG is the top priority of current BCI research, especially the feature extraction and feature classification of EEG signals based on motor imagery[2].

In recent years, the effective extraction of EEG features and improvement of classification have become an important research content of EEG signals recognition in motor imagery. At the university of New South Wales Dalin Zhang, Lina Yao was put forward based on cascade and parallel convolution recursive neural network (CRNN) recognition of EEG signals, through learn the characteristics of the original EEG signals effectively, accurately identify the human intention and instructions, in the case of a large number of dataset, have

confirmed the two model reached 98.3% of high precision, higher than that of single layer neural network for 12% of the time[3]. At the university of Middle East Technical University Yousef Rezaei Tabar and Ugur Halici proposed a brain signal classification method based on Convolutional Neural Network (CNN) and stacked autoencoders (SAE). The dataset used in the paper is the 2008 BCI IV dataset 2b, and the kappa value was 0.547, which was 9% higher than the eeg signal recognition accuracy of the winner of eeg competition[4].

New deep learning algorithms are constantly emerging, and each algorithm has its applicable fields. In the "Attention is All You Need" published by Google's machine translation team in 2017, network structures such as RNN and CNN are completely abandoned. Instead, the "Attention" mechanism is only used for machine translation tasks, and good results have been achieved. The Attention mechanism has also become a recent research hotspot[5].

In this paper, using Channel Attention Mechanism (CAM) and One Versus Rest Filter Bank Common Spatial Pattern (OVRFB-CSP) for feature extraction, using Softmax classification, and applied to imagine in EEG signals classification, the ability of the optimal feature extracting CAM, and the algorithm makes full use of OVRFB - CSP, to obtain high accuracy of classification result.

II. DATASETS

The computer environment used in the experiment is CPU, Intel(R) Core(TM) i9-9900K (3.60ghz); RAM, 32GB; Windows 10.

A. Datasets

The data used in this paper are from 2008 BCI Competition IV 2a data set [6]. In this dataset, a total of 9 subjects carried out 4 categories of motor imagery experiments, 4 categories of which were left hand, right hand, foot and tongue respectively.

Each person performed two separate period, recording 288 trials of data in each period and 72 trials in each category. In the simulation experiment, the dataset of one period is used as the training data, and the dataset of another period is used as the test data.

Twenty-five electrodes were used to record the signals, including 22 EEG channels and 3 EOG channels, as shown in Figure 1. The sampling frequency of the signal is 250Hz, and all signals are filtered by bandpass of 0.5-100Hz.

This work was supported in part by the Natural Science Foundation of Fujian Province of China under grant 2017J01117 and the Education Reform Project of Fujian Province of China under grant FBJG20190203. Fang Huijuan is the corresponding author (e-mail: fhj@hqu.edu.cn).

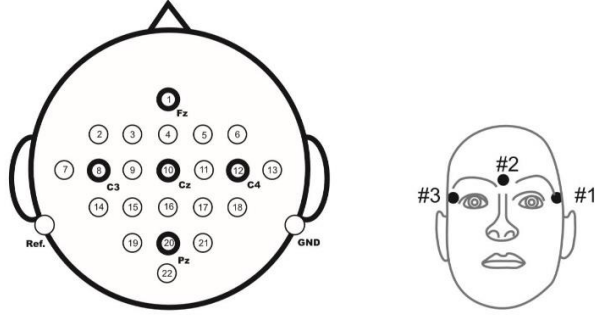


Figure 1. Electrodes position

B. Data processing

Since the experimental data of this dataset are collected continuously, EEG signals within 3s response time after stimulation should be extracted according to the experimental paradigm first, and partial NaN are found in the extracted experimental data. The number of missing values of 9 subjects is shown in Table I, in which the number of NaN of 4 and 9 is the largest. Because the existence of the NaN will affect the final algorithm implementation result, this paper adopts the method of mean interpolation to complete the missing value.

TABLE I. THE NUMBER OF MISSING VALUES (NANS) IN THE EEG DATA FOR EACH SUBJECT

Subject	Left	Right	Feet	Tongue
A01	0	0	0	0
A02	0	0	0	0
A03	0	1	0	18
A04	0	7	1499	1
A05	0	127	62	0
A06	0	23	0	1
A07	0	0	0	0
A08	23	0	187	0
A09	111	39	78	166

III. METHODS

A. Deep Learning Framework

To solve the problems of difficulty in feature extraction and low recognition rate in the process of multi-type motion imagination EEG recognition, a classification method based on deep learning framework, abbreviated as AM-OVRFB-CSP method, is proposed in this paper. This method consists of three parts: data preprocessing, feature extraction and classification recognition. The specific process is shown in Figure 2.

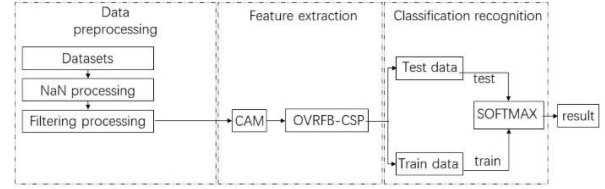


Figure 2. The overall structure of feature extraction and classification algorithm for motion-imaging EEG signals

The specific components and processing flow of each part of this algorithm are as follows:

- 1) Preprocessing part. This part firstly extracts the signal in the stimulation response area of EEG signal, removes the large interference in the data, then removes the artifact in the signal and the signal extraction of eeg signal characteristic frequency band, and finally saves the extracted data.
- 2) Feature extraction. In this part, training data is first used to construct a "one-to-many" FBCSP spatial filter, namely "OVRFB-CSP". Then the data were filtered by OVRFB-CSP, and the final EEG signal data characteristics were obtained through attention mechanism.
- 3) Feature recognition. This section is classified using the Softmax classifier.

B. Channel Attention Mechanism (CAM)

The attention mechanism is indicative of a similar degree. The closer the input is to the target, the higher the weight will be, indicating the closer the state of the input and output will be [7][8]. The principle of attention mechanism of channel feature domain is based on this idea. The information and importance of each channel will not be exactly the same, so it is necessary to reflect the degree of correlation between the features of different channels and information.

This paper will be introduced to the attention mechanism recognition study of motor imagery, constructs the Channel Attention Mechanism (CAM). The input is a preprocessed signal, the attention layer weighted the input information sequence extracted from the input data and the location of eeg channel plane, highlighting the contribution of important state information, and effectively improving the accuracy of the model evolutionary relationship. Channel Attention Mechanism as shown in Figure 3:

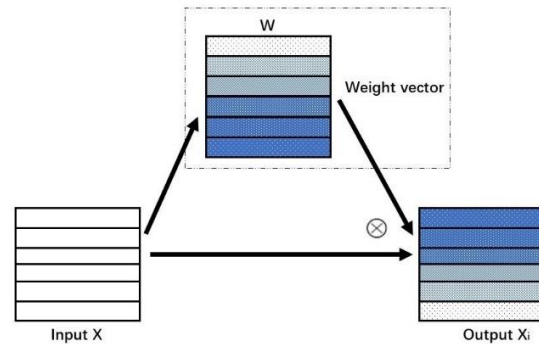


Figure 3. Channel Attention Mechanism

The CAM calculates the input data X to obtain the weight vector W corresponding to the channel. Its length is the same as the number of channels of the input data X . The element size in W indicates the degree of correlation between the data on each channel and the key information.

The weighted combination of W and each channel feature of the input data is obtained to obtain the weighted output data X_i . The different depth on X_i represents the contribution of the corresponding channel feature. The input data X is the different channel feature output from the previous layer. in (1),

$$X \in R^{N \times C} \quad (1)$$

N represents the feature vector length of each channel of eeg signal, C represents the number of channels of EEG feature vector, output data X_1 , and The process is represented in (2),

$$X_1 = W(X) \otimes X \quad (2)$$

and $\otimes$ means multiply by the elements.

The internal mechanism of channel attention mechanism is shown in Figure 4 below. The channel attention mechanism first integrates the time-order direction information of the input data, and this paper using the methods of average pooling and maximum pooling to process the input data, so as to obtain the characteristics X_{avg} and X_{max} after average pooling and maximum pooling. Then, the Multi-Layer Perceptron (MLP) is used to calculate the input X_{avg} and X_{max} to obtain two new feature vectors, and the required channel eigenweight vectors are obtained by summing the elements one by one. the number of neurons in the input layer and output layer of MLP is the same as the number of input feature channels, which is used to build the contribution degree of X_{avg} and X_{max} input channel feature information.

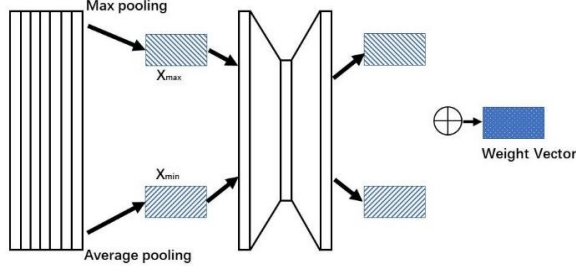


Figure 4. The internal mechanism of channel attention mechanism

CAM is represented in(3),

$$\begin{aligned} M(X) &= \varsigma \left(MLP(AvgPool(X)) \right) \\ &\quad + MLP \left(MAX(MaxPool(X)) \right) \\ &= \varsigma(W_1(W_0(X_{avg})) + W_1(W_0(X_{max}))) \end{aligned} \quad (3)$$

and ς Is the sigmoid activation function, output is limited to between 0 and 1. W_0 and W_1 are the weights of the input layer to the hidden layer and the hidden layer to the output layer respectively, X_{avg} and X_{max} represents the feature vectors obtained after average pooling and maximum pooling.

C. One Versus Rest Common Spatial Pattern(OVR-CSP)

One Versus Rest Common Spatial Pattern(OVR-CSP) treats one of the four kinds of types as a kind of type and the other three as a kind of type, so the four kinds of types are transformed into four kinds of two kinds of types[9][10]. To adopt this scheme, four spatial filters need to be constructed. Eight feature vectors were obtained from the signal through four filters, and these four sets of eigenvalues were sent into the classifier. OVRFB-CSP was obtained by combining FBCSP and OVR-CSP. The flowchart of OVRFB-CSP is shown in Figure 5:

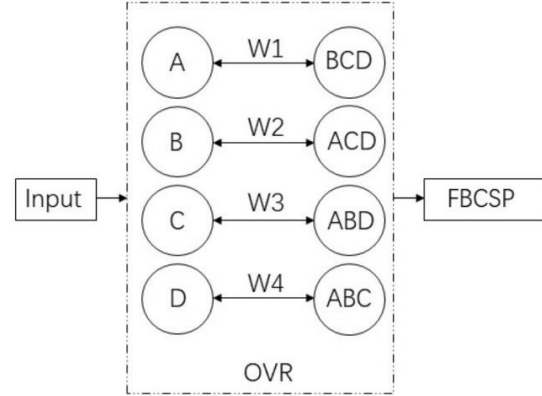


Figure 5. The flowchart of OVRFB-CSP

By using the above method to extract the eeg data from 72 single experiments (288 in total) of various kinds of motor imagery, the feature set matrix of EEG signals of left hand, right hand, foot and tongue can be obtained respectively.

D. Filter Bank Common Spatial Pattern (FBCSP)

The common spatial Pattern (CSP) is used to obtain the spatial characteristics of different types of EEG signals, and the diagonalization characteristics of the matrix are used to maximize the variance value difference between the two types of data. The specific process of FBCSP for filter Banks[11], in Figure 6:

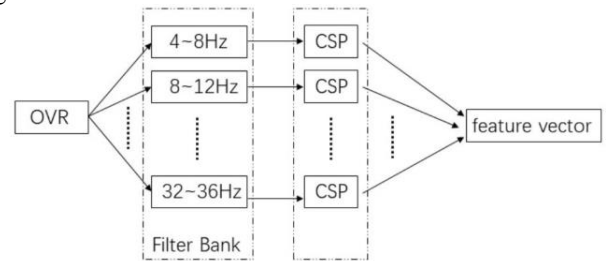


Figure 6. Flow chart of Filter Bank Common Spatial Pattern

It consists of three stages: multi-bandpass filter, spatial filter using CSP algorithm and feature selection of CSP algorithm.

IV. RESULTS

At present, the traditional recognition and classification of motion imagination mainly adopts support vector machine

(SVM) [12][13], while the recognition and classification methods based on deep learning mainly include convolutional neural network (CNN) and Recurrent Neural Network (RNN). In this study, three classification algorithms of Support Vector Machine (SVM), Convolutional Neural Network (CNN) and Attention Mechanism are used for comparative study. As can be seen from Figure 7, among the nine subjects, eight of them had the highest accuracy rate when using channel attention mechanism, and only the ninth subject had the best effect when using SVM. Compared with the other two methods, the accuracy rate improved by OVRFCSP-Attention is shown in Table II. OVRFCSP-SVM is referred to as OCS, and OVRFCSP-CNN is referred to as OCC.

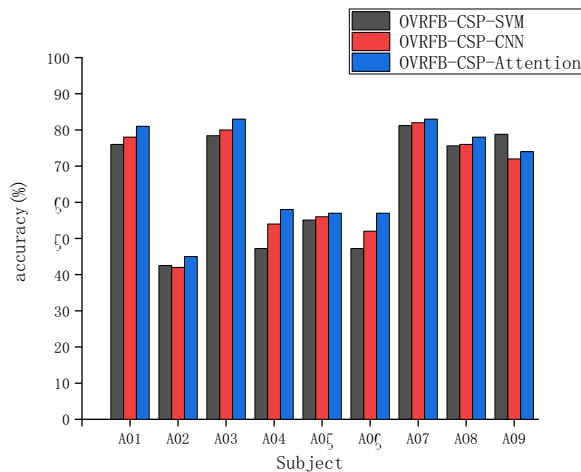


Figure 7. Comparison of three algorithms

TABLE II. OVRFCSP-ATTENTION COMPARED WITH THE OTHER TWO METHODS(%)

sub alg	A01	A02	A03	A04	A05	A06	A07	A08	A09	aver age
OCS	6.6	5.9	5.9	23	3.4	20.8	2.2	3.2	6.1	7.2
OCC	3.8	7.1	3.8	7.4	1.8	9.6	1.2	2.6	2.8	4.5

As can be seen from Table II, channel attention mechanism has a better effect on EEG signal recognition task than support vector machine and convolutional neural network. At the same time, this paper also compared the experimental results with different compression rate R parameters. It can be seen from the experimental results that different compression rate R has a certain impact on the recognition of EEG signals, and it is necessary to adjust the compression rate R accordingly for different EEG signals. For the eeg signal task in this paper, the best effect is achieved when the compression rate R is set as 8. We make an average of the accuracy rate obtained by the nine subjects under different parameters r. The experiment proves that when R is set as 8, it is 0.19%, 0.06% and 3.27% higher than when R is set as 4, 16 and 32 respectively. See figure 8.

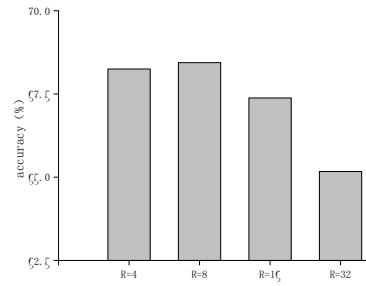


Figure 8. Different compression rates R

By comparing the brain electrical signal recognition of the existing research results and the research method of this paper, comparing the results as shown in figure 9, literature [9], respectively, using the One Versus Rest Common Spatial Pattern (OVR - CSP) and Pair Wise-CSP(PW - CSP) for feature extraction, using the SVM classification, the average accuracy rate reached 68.8% and 59.6%; Literature [14] adopted the Filter Bank Common Spatial Pattern (FBCSP) for feature extraction, and used Several Regularized Linear Discriminant Analysis (SRLDA) for classification, with average correct rate reaching 73.9%. In literature [15], Filter Bank Common Spatial Pattern (FBCSP) was adopted for feature extraction, and the Convolutional Neural Network (CNN) was used for classification, the average accuracy reached 74.1%. Literature [16] has adopted the One Versus Rest Filter Bank Common Spatial Pattern (OVRFB-CSP) and Convolutional Neural Network for feature extraction. It has a Recurrent Neural Network (RNN) for classification, whose average accuracy rate reaches 79.4%. In this paper, features were extracted by the One Versus Rest Filter Bank Common Spatial Pattern (OVRFB-CSP) and Channel Attention Mechanism (CAM), and classified by Softmax classifier. The average accuracy rate reached 68.4%. Although the overall accuracy rate of the method proposed in this paper is lower than that of the method proposed in part, it can maintain stable accuracy even when the data of the subjects is poor, such as A04 and A06.

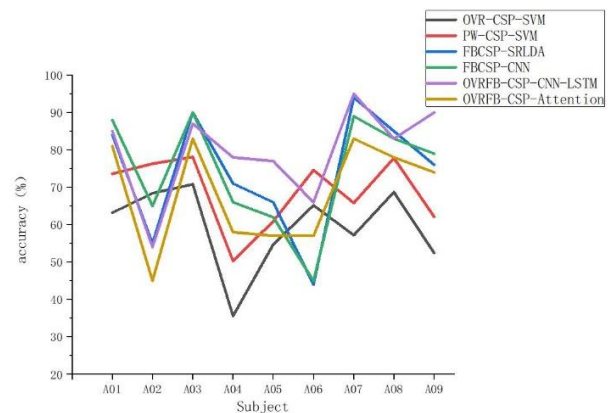


Figure 9. Comparison between the results of previous research on eeg signal recognition and those of this paper

V. CONCLUSION

In view of the difficulties in feature extraction and low recognition accuracy in the process of multi-class motion image EEG recognition, this paper proposes a multi-class EEG recognition method based on deep learning. CAM has the advantage of utilizing spatial information, CAM combined with OVRFB-CSP was used for feature extraction and Softmax classifier was used for classification. The experimental results show that CAM can fully extract the correlation information between channels in the original signal, and can extract better features, which proves the effectiveness of the proposed method. It can be seen from [9] that the effect of feature extraction using PW-CSP is better than OVR-CSP. In the future, the combination of PW-CSP with CAM will be used for research.

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